

Problem 2a: Understanding the Lagrangian (For People Who Skipped Lecture)

ECO 310 - Starting from Scratch

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1 What Even IS a Lagrangian? (The Basics)

1.1 The Problem We're Trying to Solve

Remember from Problem 1, we wanted to find the cheapest way to reach utility u^* . That's an **optimization problem**:

The Expenditure Minimization Problem

Minimize: $p_1x_1 + p_2x_2$ (total cost)

Subject to:

- $x_1 + x_2 \geq u^*$ (must reach utility u^*)
- $x_1 \geq 0$ (can't buy negative amounts)
- $x_2 \geq 0$ (can't buy negative amounts)

The challenge: You can't just minimize $p_1x_1 + p_2x_2$ by setting everything to zero, because you have **constraints**! You need $x_1 + x_2 \geq u^*$.

1.2 What's a Lagrangian?

A **Lagrangian** is a mathematical tool that combines your objective (what you're minimizing/maximizing) with your constraints into ONE big equation.

It's like a "master equation" that keeps track of everything at once.

Why do we need it? Because solving constrained optimization problems directly is hard. The Lagrangian method gives us a systematic way to find the answer.

1.3 The Magic of Lambda (λ)

The Lagrangian uses mysterious Greek letters called **Lagrange multipliers** (written as $\lambda_1, \lambda_2, \lambda_3$, etc.).

What are they?

- They're variables (like x and y) that we'll solve for

- Each λ is attached to one constraint
- They measure "how much does the objective change if we relax this constraint slightly?"

Intuition: Think of each λ as a "price" or "penalty" for violating a constraint.

2 Breaking Down THIS Specific Lagrangian

The problem gives you this Lagrangian:

$$\mathcal{L} = p_1x_1 + p_2x_2 + \lambda_1(-x_1) + \lambda_2(-x_2) + \lambda_3(x_1 + x_2 - u^*) \quad (1)$$

Let's break it down piece by piece.

2.1 Part 1: The Objective Function

$$\boxed{p_1x_1 + p_2x_2} \quad (2)$$

What it is: This is what we're trying to **minimize** - the total cost.

Why it's here: The Lagrangian always starts with the objective function (the thing you're optimizing).

Example:

- If coffee costs \$3 and tea costs \$5
- And you buy 10 coffees and 0 teas
- Cost = $3(10) + 5(0) = \$30$

2.2 Part 2: Nonnegativity Constraint on Good 1

$$\boxed{\lambda_1(-x_1)} \quad (3)$$

Original constraint: $x_1 \geq 0$ (can't buy negative coffee)

Why it's written as $-x_1$:

- We need to rewrite $x_1 \geq 0$ as $-x_1 \leq 0$ (multiply both sides by -1)
- Or equivalently: $0 - x_1 \leq 0$
- The Lagrangian uses the form: constraint ≤ 0

What λ_1 means:

- λ_1 is the "shadow price" of the nonnegativity constraint on good 1
- If $\lambda_1 > 0$: the constraint is **binding** (you're at $x_1 = 0$)
- If $\lambda_1 = 0$: the constraint is **not binding** (you're buying $x_1 > 0$)

Intuition: If you're forced to buy exactly 0 of good 1 (hitting the constraint), λ_1 tells you "how much would your cost decrease if you could buy a tiny bit?"

2.3 Part 3: Nonnegativity Constraint on Good 2

$$\boxed{\lambda_2(-x_2)} \tag{4}$$

Original constraint: $x_2 \geq 0$ (can't buy negative tea)

Why it's written as $-x_2$: Same logic as above - rewrite $x_2 \geq 0$ as $-x_2 \leq 0$

What λ_2 means:

- λ_2 is the "shadow price" of the nonnegativity constraint on good 2
- If $\lambda_2 > 0$: you're at $x_2 = 0$ (constraint is binding)
- If $\lambda_2 = 0$: you're buying $x_2 > 0$ (constraint is not binding)

2.4 Part 4: Utility Constraint

$$\boxed{\lambda_3(x_1 + x_2 - u^*)} \tag{5}$$

Original constraint: $x_1 + x_2 \geq u^*$ (must reach utility level u^*)

Why it's written as $(x_1 + x_2 - u^*)$:

- Rewrite $x_1 + x_2 \geq u^*$ as $x_1 + x_2 - u^* \geq 0$
- For a $\geq$ constraint, we use the form: constraint ≥ 0

What λ_3 means:

- λ_3 is the "shadow price" of utility
- It measures: "how much would minimum cost increase if we required one more unit of utility?"
- This is actually the **marginal cost of utility**
- For expenditure minimization, $\lambda_3 > 0$ always (we're always hitting the utility constraint)

Intuition: If you need to be happier (higher u^*), you need to spend more. λ_3 tells you exactly how much more per unit of utility.

3 Why This Form? The General Pattern

3.1 General Rules for Building Lagrangians

For a minimization problem:

Lagrangian Recipe

Start with: Objective function (what you're minimizing)

For each constraint of type $g(x) \leq 0$:

- Add: $+\lambda_i \cdot g(x)$
- Where $\lambda_i \geq 0$

For each constraint of type $h(x) \geq 0$:

- Add: $+\mu_j \cdot h(x)$
- Where $\mu_j \geq 0$

3.2 Applying to Our Problem

Objective: Minimize $p_1x_1 + p_2x_2$

Constraints:

1. $x_1 \geq 0 \rightarrow$ rewrite as $-x_1 \leq 0 \rightarrow$ add $\lambda_1(-x_1)$
2. $x_2 \geq 0 \rightarrow$ rewrite as $-x_2 \leq 0 \rightarrow$ add $\lambda_2(-x_2)$
3. $x_1 + x_2 \geq u^* \rightarrow$ rewrite as $(x_1 + x_2 - u^*) \geq 0 \rightarrow$ add $\lambda_3(x_1 + x_2 - u^*)$

Result:

$$\mathcal{L} = p_1x_1 + p_2x_2 + \lambda_1(-x_1) + \lambda_2(-x_2) + \lambda_3(x_1 + x_2 - u^*)$$

4 Complete Explanation (Part 2a Answer)

Answer to Problem 2a

This Lagrangian represents the expenditure minimization problem for perfect substitutes.
Each part represents:

1. $p_1x_1 + p_2x_2$: The objective function - total expenditure we're minimizing
2. $\lambda_1(-x_1)$: Nonnegativity constraint on good 1 ($x_1 \geq 0$).
 - $\lambda_1 > 0$ if we buy exactly 0 of good 1 (constraint binds)
 - $\lambda_1 = 0$ if we buy some good 1 (constraint doesn't bind)
3. $\lambda_2(-x_2)$: Nonnegativity constraint on good 2 ($x_2 \geq 0$).
 - $\lambda_2 > 0$ if we buy exactly 0 of good 2 (constraint binds)
 - $\lambda_2 = 0$ if we buy some good 2 (constraint doesn't bind)
4. $\lambda_3(x_1 + x_2 - u^*)$: Utility constraint ($x_1 + x_2 \geq u^*$).
 - Ensures we reach utility level u^*
 - λ_3 represents the marginal cost of utility (how much extra spending is needed for one more unit of utility)
 - This constraint always binds (we choose exactly $x_1 + x_2 = u^*$ to minimize cost)

Together: The Lagrangian combines the objective with all constraints into one equation that can be optimized using calculus (taking derivatives).

5 Concrete Example: Coffee and Tea

Let's see this in action with our coffee/tea example.

Setup:

- Coffee (good 1) costs \$3: $p_1 = 3$
- Tea (good 2) costs \$5: $p_2 = 5$
- Want utility 10: $u^* = 10$

The Lagrangian:

$$\begin{aligned}\mathcal{L} &= 3x_1 + 5x_2 + \lambda_1(-x_1) + \lambda_2(-x_2) + \lambda_3(x_1 + x_2 - 10) \\ &= 3x_1 + 5x_2 - \lambda_1x_1 - \lambda_2x_2 + \lambda_3(x_1 + x_2 - 10)\end{aligned}$$

What we expect:

- Coffee is cheaper, so buy only coffee: $x_1 = 10, x_2 = 0$
- Since $x_1 = 10 > 0$, the constraint $x_1 \geq 0$ is NOT binding $\rightarrow \lambda_1 = 0$

- Since $x_2 = 0$, the constraint $x_2 \geq 0$ IS binding $\rightarrow \lambda_2 > 0$
- Since we need exactly utility 10, the utility constraint IS binding $\rightarrow \lambda_3 > 0$

Economic interpretation:

- $\lambda_1 = 0$: We're not constrained by $x_1 \geq 0$ (we happily buy coffee)
- $\lambda_2 > 0$: We'd like to buy negative tea if we could (to get a refund!), but we can't, so this constraint bites
- $\lambda_3 > 0$: If we needed higher utility, we'd have to spend more - specifically, $\lambda_3 = 3$ (the cost per unit of utility = price of coffee)

6 Why Do We Need the Lagrangian?

Can't we just solve it directly? For perfect substitutes, yes! But for more complex utility functions (Cobb-Douglas, CES, etc.), you can't easily solve by inspection.

What the Lagrangian gives us:

1. A systematic method that works for ANY utility function
2. First-order conditions (derivatives = 0) that we can solve
3. Information about which constraints are binding (from the λ values)
4. Economic interpretations (shadow prices)

7 The Big Picture

Understanding Lagrangians

What it is: A tool for solving constrained optimization problems

How it works:

1. Start with objective function
2. Add each constraint, multiplied by a λ (Lagrange multiplier)
3. Take derivatives with respect to all variables ($x_1, x_2, \lambda_1, \lambda_2, \lambda_3$)
4. Set derivatives equal to zero
5. Solve the system of equations

What you get:

- The optimal values of x_1, x_2 (compensated demand)
- The shadow prices λ_i (marginal values of relaxing each constraint)
- Which constraints are binding (where $\lambda_i > 0$)

For Problem 2: You're using this Lagrangian to derive the same answer you got in Problem 1a, but now using a formal mathematical method rather than intuition.

8 Connection to Part (b) and (c)

Part 2b: Will ask you to write the optimality conditions (derivatives of $\mathcal{L}$ set equal to zero) and solve them.

Part 2c: Will show that when you guess wrong about which constraints bind, you get an invalid solution (like negative λ values or violated constraints).

Key lesson: For perfect substitutes, exactly one nonnegativity constraint will bind (you buy only one good), and the utility constraint always binds.

9 IMPORTANT: Two Sign Conventions (Which One Are You Using?)

There are TWO common ways to write the Lagrangian for minimization problems. They give the same answer but look different!

9.1 Convention 1: Positive Objective (What I Used Above)

$$\mathcal{L} = p_1x_1 + p_2x_2 + \lambda_1(-x_1) + \lambda_2(-x_2) + \lambda_3(x_1 + x_2 - u^*) \quad (6)$$

Expanding:

$$\mathcal{L} = p_1x_1 + p_2x_2 - \lambda_1x_1 - \lambda_2x_2 + \lambda_3(x_1 + x_2 - u^*)$$

This convention: Keeps the objective positive, adds constraints with their natural signs.

9.2 Convention 2: NEGATIVE Objective (Probably Your Homework!)

$$\mathcal{L} = -p_1x_1 - p_2x_2 + \lambda_1x_1 + \lambda_2x_2 + \lambda_3(x_1 + x_2 - u^*) \quad (7)$$

Or factored:

$$\mathcal{L} = -(p_1x_1 + p_2x_2) + \lambda_1x_1 + \lambda_2x_2 + \lambda_3(x_1 + x_2 - u^*)$$

This convention: Negates the objective function, turning minimization into maximization.

Why the Negative Sign?

Mathematical trick: Minimizing $f(x)$ is the same as maximizing $-f(x)$!

So:

- **Minimize** cost $(p_1x_1 + p_2x_2)$
- = **Maximize** negative cost $-(p_1x_1 + p_2x_2)$

Some textbooks prefer this because it makes all optimization problems look like maximization.

9.3 How to Tell Which Convention Your Problem Uses

Check the Lagrangian given in the problem!

If you see: $\mathcal{L} = p_1x_1 + p_2x_2 + \dots \rightarrow$ Convention 1 (positive objective)

If you see: $\mathcal{L} = -p_1x_1 - p_2x_2 + \dots \rightarrow$ Convention 2 (negative objective)

9.4 Breaking Down Convention 2 (Negative Objective)

Let's explain each piece of:

$$\mathcal{L} = -(p_1x_1 + p_2x_2) + \lambda_1x_1 + \lambda_2x_2 + \lambda_3(x_1 + x_2 - u^*)$$

9.4.1 Part 1: $-(p_1x_1 + p_2x_2)$

What it is: Negative of total cost

Why negative? We're converting "minimize cost" into "maximize negative cost"

Interpretation: This is what we're maximizing (which is equivalent to minimizing cost)

9.4.2 Part 2: $+\lambda_1x_1$

Original constraint: $x_1 \geq 0$ (nonnegativity on good 1)

Why positive? With the negative objective convention:

- For $\geq$ constraints, add $+\lambda \cdot (\text{constraint})$

- So: $x_1 \geq 0$ becomes $+\lambda_1x_1$

What λ_1 means: (Same as before!)

- $\lambda_1 > 0$ if $x_1 = 0$ (constraint binds)
- $\lambda_1 = 0$ if $x_1 > 0$ (constraint doesn't bind)

9.4.3 Part 3: $+\lambda_2x_2$

Original constraint: $x_2 \geq 0$ (nonnegativity on good 2)

Why positive? Same logic: $x_2 \geq 0$ becomes $+\lambda_2x_2$

What λ_2 means:

- $\lambda_2 > 0$ if $x_2 = 0$ (constraint binds)
- $\lambda_2 = 0$ if $x_2 > 0$ (constraint doesn't bind)

9.4.4 Part 4: $+\lambda_3(x_1 + x_2 - u^*)$

Original constraint: $x_1 + x_2 \geq u^*$ (utility constraint)

Same as Convention 1! This part doesn't change between conventions.

What λ_3 means: Marginal cost of utility (always positive, constraint always binds)

9.5 Side-by-Side Comparison

Part	Convention 1	Convention 2
Objective	$+p_1x_1 + p_2x_2$	$-p_1x_1 - p_2x_2$
$x_1 \geq 0$	$-\lambda_1x_1$	$+\lambda_1x_1$
$x_2 \geq 0$	$-\lambda_2x_2$	$+\lambda_2x_2$
$x_1 + x_2 \geq u^*$	$+\lambda_3(x_1 + x_2 - u^*)$	$+\lambda_3(x_1 + x_2 - u^*)$

Key point: The signs on the nonnegativity constraints flip, but the utility constraint stays the same!

9.6 Complete Explanation for Convention 2

Answer to Problem 2a (Convention 2)

This Lagrangian represents the expenditure minimization problem written as a maximization problem.

Each part represents:

1. $-(p_1x_1 + p_2x_2)$: Negative of expenditure - we maximize this (= minimize expenditure)
2. λ_1x_1 : Nonnegativity constraint on good 1 ($x_1 \geq 0$).
 - $\lambda_1 > 0$ if we buy exactly 0 of good 1 (constraint binds)
 - $\lambda_1 = 0$ if we buy some good 1 (constraint doesn't bind)
3. λ_2x_2 : Nonnegativity constraint on good 2 ($x_2 \geq 0$).
 - $\lambda_2 > 0$ if we buy exactly 0 of good 2 (constraint binds)
 - $\lambda_2 = 0$ if we buy some good 2 (constraint doesn't bind)
4. $\lambda_3(x_1 + x_2 - u^*)$: Utility constraint ($x_1 + x_2 \geq u^*$).
 - Ensures we reach utility level u^*
 - λ_3 represents the marginal cost of utility
 - This constraint always binds ($x_1 + x_2 = u^*$)

Economic interpretation is identical to Convention 1! Only the signs differ.

9.7 Which Convention Should You Use?

USE THE ONE YOUR HOMEWORK GIVES YOU!

- Look at the Lagrangian written in the problem
- Copy the sign convention exactly
- The economic interpretation is the same either way
- The optimal solution (x_1^*, x_2^*) is the same either way

Bottom Line

If your homework shows:

$$\mathcal{L} = -p_1x_1 - p_2x_2 + \lambda_1x_1 + \lambda_2x_2 + \lambda_3(x_1 + x_2 - u^*)$$

Then use Convention 2. Everything I explained still applies - just flip the signs on the objective and nonnegativity constraints!

The math works out to the same answer either way.